# ZIYANG CHEN

Ann Arbor, MI ◇ czyang@umich.edu ◇ Webpage

## EDUCATION

**University of Michigan, Ann Arbor** Jan. 2022 -
Ph.D. in Electrical & Computer Engineering | Advisor: *Andrew Owens*

**University of Michigan, Ann Arbor** Sept. 2019 - May 2021
Master of Science in Engineering, Computer Vision
Overall GPA: **4.0/4.0** | Advisor: *Andrew Owens*

**Shanghai Jiao Tong University, UM-SJTU JI** Sept. 2015 - Aug. 2019
Bachelor of Science in Engineering
Major: Electrical & Computer Engineering | Minor: Entrepreneurship
GPA: **3.60/4.0** with honors & *Graduated magna cum laude*

## PUBLICATIONS

- *Ziyang Chen, Shengyi Qian and Andrew Owens.* Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation. (In submission) (Project site)
- *Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.* Conditional Generation of Audio from Video via Foley Analogies. (CVPR 2023)
- *Chao Feng, Ziyang Chen and Andrew Owens.* Self-Supervised Video Forensics by Audio-Visual Anomaly Detection. (CVPR 2023) (Project site)
- *Ziyang Chen, David F. Fouhey and Andrew Owens.* Sound Localization by Self-Supervised Time Delay Estimation. (ECCV 2022) (Project site)
- *Xixi Hu*, Ziyang Chen* and Andrew Owens.* Mix and Localize: Localizing Sound Sources in Mixtures. (CVPR 2022) (Project site)
- *Ziyang Chen*, Xixi Hu* and Andrew Owens.* Structure from Silence: Learning Scene Structure from Ambient Sound. (CoRL 2021, Oral) (Project site)

## RESEARCH EXPERIENCE

**Sound Localization from Motion** Aug. 2022 - Present
*Research Associate, University of Michigan* *Advisor: Andrew Owens*
- Proposed a self-supervised method to jointly learn to estimate sound source direction and camera pose
- Proposed an audio-visual representation learning through a self-supervised cross-view binauralization task *(In submission)*

**Video Forensics by Audio-Visual Anomaly Detection** Jan. 2021 - Mar. 2023
*Research Associate, University of Michigan* *Advisor: Andrew Owens*
- Proposed a zero-shot audio-visual anomaly detection method by leveraging the synchronization between audio and frames in the real videos *(CVPR 2023)*

**Conditional Foley Generation** June 2021 - Mar. 2023
*Research Associate, University of Michigan* *Advisor: Bryan Russell, Justin Salamon, Andrew Owens*
- Proposed a self-supervised conditional Foley model that creates audio tracks for silent videos conditioning on given videos and sounds *(CVPR 2023)*

**Sound Localization by Self-Supervised Time Delay Estimation** June 2021 - June 2022
*Research Associate, University of Michigan* *Advisor: David F. Fouhey, Andrew Owens*
- Proposed self-supervised methods to learn interaural time difference from Internet stereo or mono audio

- Proposed a novel visually-guided time delay estimation task: localizing a speaker in a multi-speaker recording
- Evaluated on simulated environments with metrically accurate ground truth and on internet videos with directional judgements achieving significantly better results *(ECCV 2022)*

**Mix and Localize** July 2020 - March 2022
*Research Associate, University of Michigan* *Advisor: Andrew Owens*

- Proposed a simple self-supervised audio-visual method using contrastive random walk with cycle-consistency constraints and applied learned representations to sound source localization task
- Implemented state-of-the-art sound localization methods as baselines
- Evaluated our method on human speech and musical instruments datasets and achieved significantly better qualitative and quantitative results *(CVPR 2022)*

**Structure from Silence** Jan. 2020 - May 2020, Apr. 2021 - Sep. 2021
*Research Associate, University of Michigan* *Advisor: Andrew Owens*

- Collected datasets with ambient sound and paired RGB-D videos of the indoor environment
- Demonstrated that the ambient sound could infer geometric properties with experiments and proposed a new audio-visual learning method to learn exploit structural representation
- Apply learned models to robots and navigate it via ambient sound only in the real world *(CoRL 2021)*

## WORK EXPERIENCE

**Research Assistant, *University of Michigan, Ann Arbor*** Jan. 2020 - Present

- Topics: *audio-visual representation learning*, *self-supervised learning*

**Teaching Assistant, *Shanghai Jiao Tong University*** Sept. 2018 - Aug. 2019

- Courses: *Digital Integrated Circuits*, *Electronic Circuits*
- Conducted experiments, held office hours and taught recitation classes every week

**Research Engineer Intern, *FreeSense Image Inc., Shanghai*** Mar. 2018 - May 2018

- Implemented an auxiliary system for a project of Detection of Car LCD Screen and applied it into industrial production lines
- Adopted state-of-the-art recognition DNNs to improve the accuracy on defect detection tasks

## SERVICE

**Reviewer**

- WACV 2023, CVPR 2023

## AWARDS & SCHOLARSHIPS

- **Lums Fellowship**, *University of Michigan, Ann Arbor* 2019-2021
- **Outstanding Graduate - Class of 2019**, *Shanghai Jiao Tong University* 2019
- **Academic Excellence Scholarship**, *Shanghai Jiao Tong University* 2016-2018
- **Dean's List**, *Shanghai Jiao Tong University, SJTU-UM JI* 2015-2018